

CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2011

BIOLOGY
Higher 2

Paper 3

**Monday
22 August 2011**

2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

FOR EXAMINER'S USE	
1	/15
2	/14
3	/11
4	/12
5	/20
TOTAL	/72

At the end of examination,

1. fasten all your work securely together;

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 13 printed pages and 1 blank page

Answer all questions in the space provided

Question 1

There are a few methods to synthesize recombinant insulin. One of the methods involves cloning the human insulin cDNA into the cloning vector pUC19 and transforming the clone into *E. coli*.

The modified *E. coli* plasmid vector, pUC19 developed by Fermentas is shown in **Fig. 1.1**. The pUC19 plasmid vector is small (2686bp) with high copy number. The plasmid contains 1) the pMB1 replicon rep (also known as the *Ori* site) responsible for the replication of plasmid, 2) *bla* gene encoding for beta-lactamase that confers resistance to ampicillin and 3) a region of *E. coli* lac operon containing CAP protein binding site, promoter Plac, lac repressor binding site and 5'-terminal part of the lacZ gene encoding the N-terminal fragment of beta-galactosidase. The map shows restriction enzymes that cut pUC19 once.

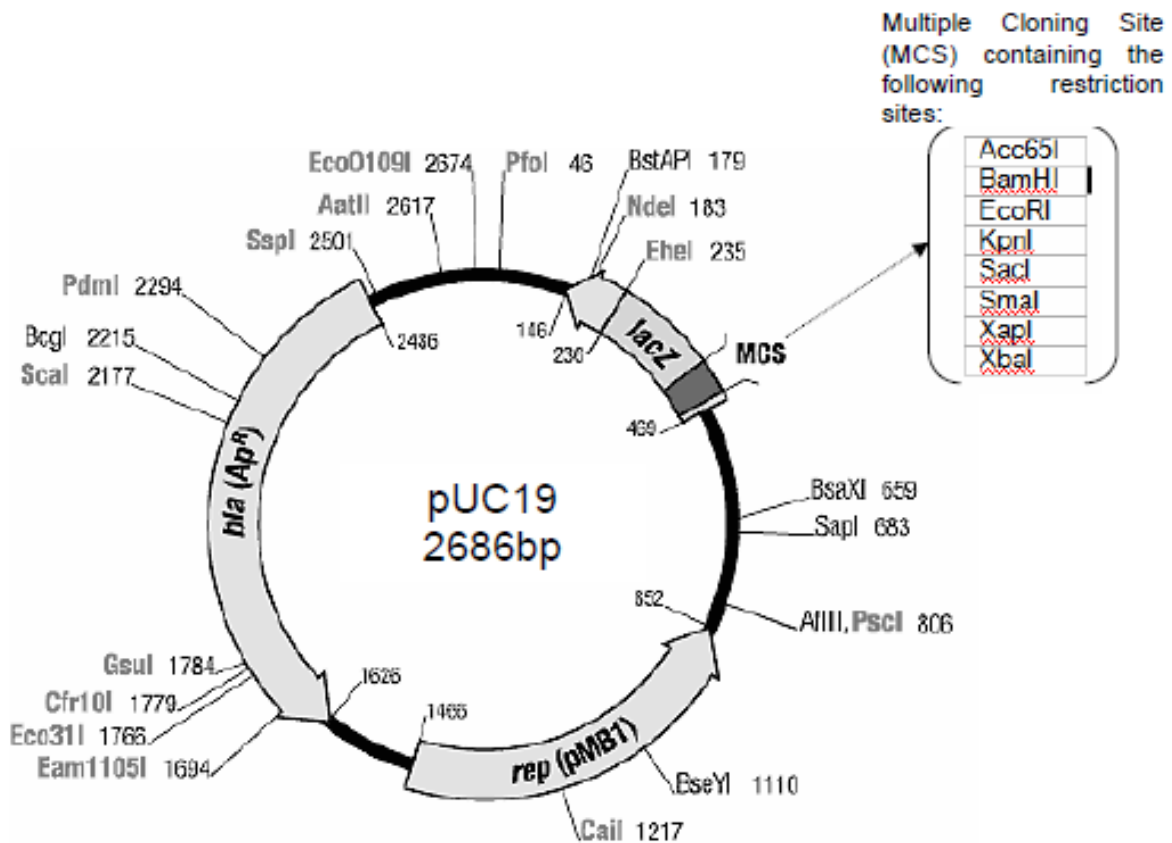


Fig. 1.1

- (a) With reference to **Fig. 1.1**, by means of a **labelled arrow**, indicate on the diagram the most suitable position to insert the fragment of human insulin cDNA. [1]

- (b) Using the information given in **Table 1.1**, select a restriction enzyme that will generate a blunt end and also allow for subsequent visual identification of successful recombinant colonies.

(i) State your choice of restriction enzyme:[1]

(ii) Explain your choice of restriction enzyme. [1]

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Restriction enzyme	Specific recognition sequence
Acc65I	5'-G [^] G T A C C-3' 3'-C C A T G [^] G-5'
BamHI	5'-G [^] G A T C C-3' 3'-C C T A G [^] G-5'
EcoRI	5'-G [^] A A T T C-3' 3'-C T T A A [^] G-5'
EheI	5'-T [^] C T A G A-3' 3'-A G A T C [^] T-5'
KpnI	5'-G G T A C [^] C-3' 3'-C [^] C A T G G-5'
Pdml	5'-G A A N N [^] N T T C-3' 3'-C T T N N [^] N A A G-5'
Scal	5'-A G T [^] A C T-3' 3'-T C A [^] T G A-5'
SmaI	5'-C C C [^] G G G-3' 3'-G G G [^] C C C-5'

[^] indicates where the restriction enzyme cuts

Table 1.1

- (c) Some restriction enzymes generate sticky ends, while others generate blunt ends. Explain why it is more efficient to clone DNA fragments with sticky ends than DNA fragments with blunt ends. [1]

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- (d) Describe how a set of blunt ended DNA fragment could still be efficiently cloned into a plasmid vector. [3]

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- (e) Outline the principle of insertional inactivation of the *lacZ* gene. [4]

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After transforming the clones into the *E. coli* cells, the cells were plated onto appropriate mediums. Positive results were observed from the morphologies of the grown colonies.

The colonies were harvested and purified via affinity chromatography. A portion of the purified extract was further analysed. Results of the analysis revealed that functional insulin was not present in the extract.

- (f) Suggest two reasons why functional insulin was not expressed in the recombinant *E. coli* cells. [4]

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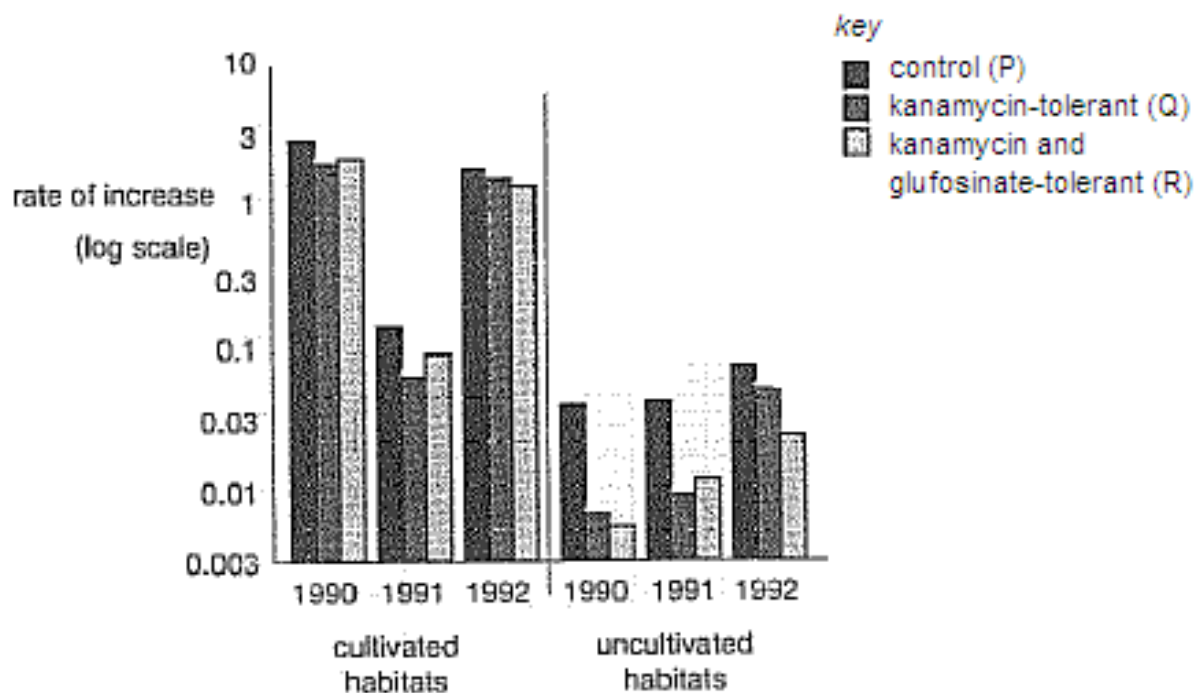
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[Total: 15 marks]

Question 2

An experiment was carried out over three years to compare the population growth of normal and genetically engineered oilseed rape plants. Invasiveness was defined as the rate of population increase from one year to the next. Populations in which the rate of increase is greater than 1 would be expected to increase in abundance.

Three different genetic lines (**P**, **Q** and **R**) were compared: non-genetic engineered oilseed rape (**P**) and two genetically engineered versions of the same cultivar. One of these (**Q**) was tolerant of the antibiotic *kanamycin* and the other (**R**) tolerant of both **kanamycin** and the herbicide *glufosinate*. Neither antibiotic nor herbicide was used in the investigation. The rate of population increase of the three genetic lines grown in cultivated and uncultivated habitats are shown in **Fig. 2.1**.



ACKNOWLEDGEMENT: Crawley et al, Nature 363 620 – 623, 17 June 1993

Fig. 2.1

- (a) Briefly explain how plants are propagated by cloning from tissue culture to obtain the cultivar. [4]

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- (b) Tolerance to the antibiotic kanamycin is not expected to give an advantage to the oilseed rape plants. Suggest why this genetic line was included in the investigation. [1]

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- (c) With reference to **Fig. 2.1**, compare the invasiveness of non-engineered and genetically engineered oilseed rape plants in cultivated and uncultivated habitats.

- (i) Cultivated habitats: [2]

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- (ii) Uncultivated habitats: [2]

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- (iii) Suggest an explanation for the observed data. [1]

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- (d) Based on the findings of this investigation,

- (i) evaluate whether the concern about transgenic crop plants becoming agricultural weeds is justified. [1]

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- (ii) describe 1 potential benefit and 1 hazard of such genetically engineered oilseed rape plants. [2]

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- (e) Explain why plants of the same family as the oilseed rape are at greatest risk of picking up the foreign gene from the genetically engineered plants compared to plants from a different family. [1]

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[Total: 14 marks]

Question 3

A new brightly-coloured mouse was discovered in Russia and it was given the name *Mus colores*. **Fig. 3.1** below shows two flowcharts that illustrate how blue and green cells were isolated from different parts of this mouse, and each cell was cultured individually.

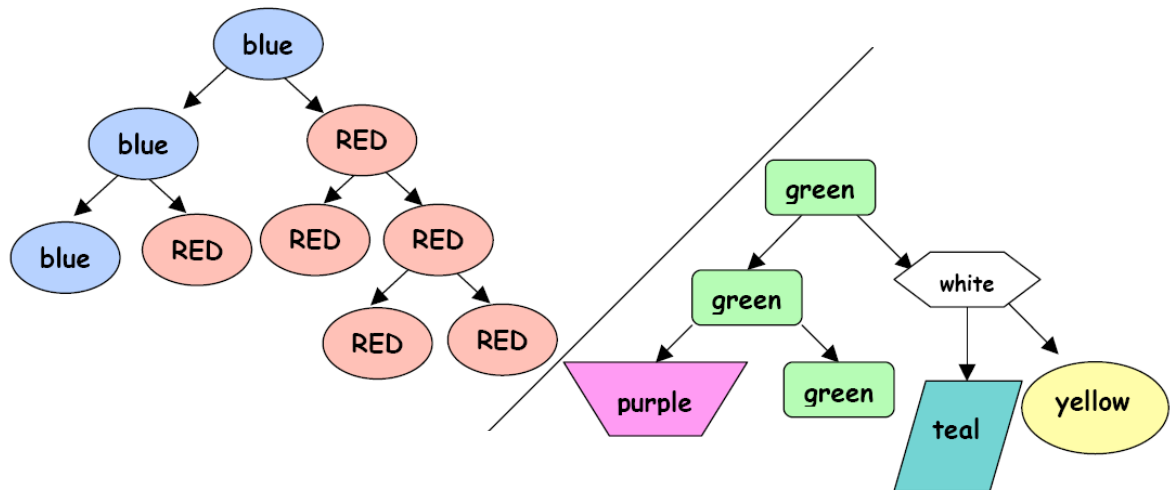


Fig. 3.1

(a) With reference to **Fig. 3.1**,

(i) are the blue cells stem cells? Give a reason for your answer. [1]

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(ii) given that the green cell is a stem cell, explain why green cells can give rise to differently coloured cells. [2]

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Severe combined immunodeficiency disorder (SCID) is an inherited autosomal recessive disorder.

(b) Discuss **one** ethical and **one** social concern regarding the use of gene therapy. [2]

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A trial was carried out on a ADA-SCID patient of age 7 months. Hematopoietic stem cells (HSCs) were collected from the patient's bone marrow, transduced with retroviral vector containing the ADA gene, and infused back. The patient received 8.6×10^6 HSCs per kg body weight, containing 25% successfully transformed cells in culture.

Figure 3.2A below shows the number of total lymphocyte counts (solid circles) (left ordinate) and white blood cell counts (open circles) (right ordinate) in the peripheral blood (PB) for this patient.

DNA was also isolated and analyzed for proportion of transformed cells. **Figure 3.2B** shows the percent of transformed cells in the following sorted cell populations: T cells (open circles) and NK cells (solid circles).

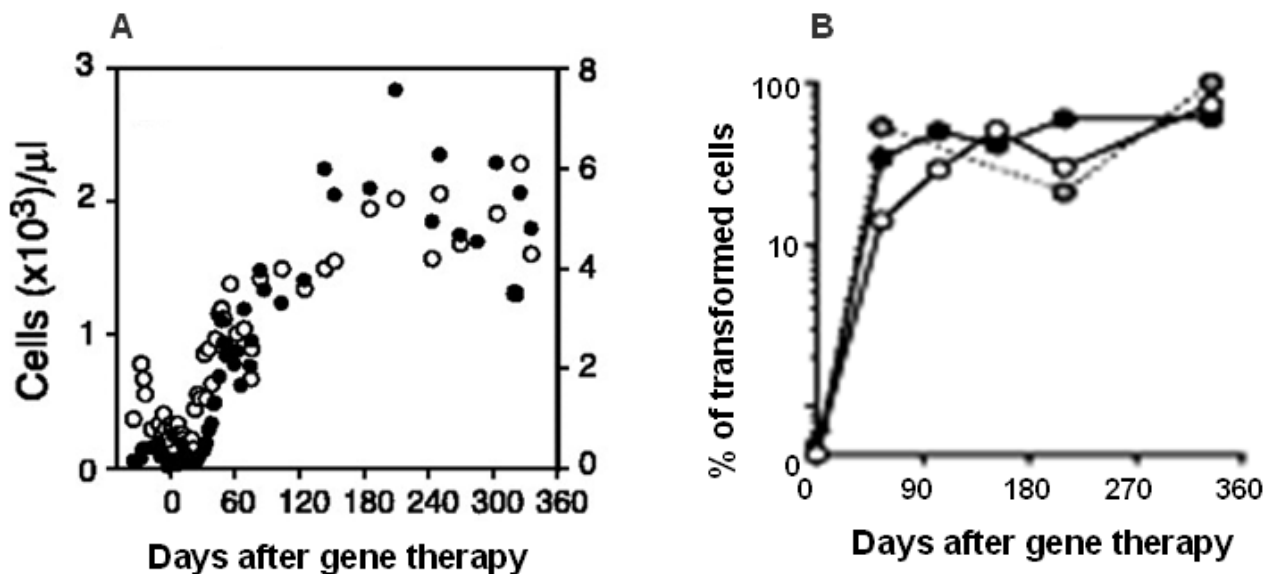


Figure 3.2

- (c) Suggest why a retroviral vector was chosen for use in this trial. [2]

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- (d) Using evidence from **Figure 3.2**, explain why haematopoietic stem cells in particular are used for gene therapy. [4]

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[Total: 11 marks]

Planning question

Question 4

One of Africa's most famous extinct animals, the quagga, was a subspecies of the plains zebra, which was once found in great numbers in South Africa's Cape Province and the southern part of the Orange Free State.

It was distinguished from other zebras by having the usual vivid marks on the front part of the body only. In the mid-section, the stripes faded and the dark, inter-stripe spaces became wider, and the hindquarters were a plain brown.

Hunters in Africa have caught an animal which resembles a quagga. However, they could not be sure.

Plan an experiment to investigate if this animal which the hunters caught is a descendant of the quagga.

Your planning must be based on the assumption that you have been provided with the following equipment and materials

- Tissue samples from museum specimens of a real quagga and from the animal that the hunters caught
- Pestle and mortar
- DNA extraction buffer solution
- Glass rods
- Microcentrifuge tubes
- Centrifuge
- Restriction enzyme
- Agarose or polyacrylamide gel plate
- Suitable source of electrical current
- Radioactive probe
- Nitrocellulose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include

- An explanation of the theory to support your practical procedure
- A description of the method used, including the scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analyzed including how the origin of the organism can be determined

[Total: 12]

Free-response question [20 marks]

Write your answer on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 5

- (a) Describe the different types of stem cells and using named examples, explain their normal functions in a living organism. [6]
- (b) Describe how cystic fibrosis can be cured using gene therapy and the technical challenges scientists still face when designing such a cure. [8]
- (c) With reference to a named example, explain how RFLP can be used in disease detection. [6]

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